Lab Seminar Presentation 1

Background

- Ammonia, hydrogen combustion important because no carbon emission
- reaction mechanisms required to study fuels at various combustion parameters
    - reduced mechanisms req for applications in cfd
    - req to predict flame properties eg. lbv, emissions, idt, etc.

Abe-san's reaction mechanism

Validating reaction mechanism - Completed

- data from experiments collated to check if reaction mechanism models real flame ppts

LBV

- nh3/h2/air:
    - han 2019
    - wang 2020
    - shrestha 2021
    - wang 2021
    - zhou 2023
- very good alignment with experimental data for most conditions
- lacking exp data for high pressure conds
- ch4/nh3/air:
    - Han 2019
    - Ling 2026
    - Okafor 2019
- models well for p=1atm cases
    - okafor 2019 exp values inconsistent with other at same conds
- slight over prediction, trend maintained
- lacking exp data for high pressure conds

Validating reaction mechanism - Incomplete

- NH3, NO, CO data exp and numerical
    - numerical use plug flow reactor
- OH, NO, NH specie data exp and numerical
    - numerical use stagnation flame model